S Lp Kin trúc (Architecture Class Diagram - S rút gn)

Project: DuAn_BanHangVaCongNoKhachHang
(Tham chiu UseCases, CoreBusiness, Plugins ti Program
cs)
«Razor Component»
TaoHoaDonPage
-ITaoHoaDonBanChiuUseCase TaoHoaDonBanChiuUseCase
-IHoaDonNhapStateService HoaDonNhapStateService
+LuuHoaDonAsync() : Task
«Razor Component»
TaoThanhToanPage
-ITaoThanhToanUseCase TaoThanhToanUseCase
-IXemTruocPhanBoFifoUseCase XemTruocPhanBoFifoUseCase
+LuuThanhToanAsync() : Task
«Razor Component»
BaoCaoTuoiNoPage
-IXemBaoCaoTuoiNoUseCase XemBaoCaoTuoiNoUseCase
-IXuatBaoCaoTuoiNoExcelUseCase XuatBaoCaoTuoiNoExcelUseCase
+XuatExcelAsync() : Task
«CompositionRoot»
Program
note: ng ký liên kt gia Interfaces và Plugins
+Main(string[] args) : void
inject Scoped state
inject interface
inject interface
inject interface
ng ký DI
ng ký DI
ng ký DI
ng ký DI
Project: BanHang
CoreBusiness
(0 ProjectReference - c lp tuyt i)
KhachHang
+int KhachHangId [PK]
+string MaKhachHang
+string TenKhachHang
+string? SoDienThoai
+string? DiaChi
+decimal HanMucCongNo
+decimal SoDuCongNo «tng hp khi truy vn»
+bool TrangThai
Thuc tính tính toán C# (không lu ct tnh SQL)
+decimal ConDuocMuaNo «computed C#»
+bool DaVuotHanMuc «computed C#»
Phng thc nghip v
+bool KiemTraVuotHanMucKhiMua(decimal soTienDonHangMoi)
SanPham
+int SanPhamId [PK]
+string MaSanPham
+string TenSanPham
+string DonViTinh
+decimal DonGia
+int SoLuongTon
+bool TrangThai
Phng thc nghip v
+bool CoDuTonKho(int soLuongCan)
HoaDon
+int HoaDonId [PK]
+string MaHoaDon
+int KhachHangId [FK]
+DateTime NgayLap
+DateTime HanThanhToan
+decimal TongTien
+decimal DaThanhToan
+TrangThaiHoaDon TrangThai
Ct tính toán SQL PERSISTED
+decimal ConLai «computed SQL PERSISTED»
Phng thc nghip v
+CapNhatTongTien() : void
+CapNhatTrangThai() : void
ChiTietHoaDon
+int ChiTietHoaDonId [PK]
+int HoaDonId [FK]
+int SanPhamId [FK]
+string TenSanPham
+string DonViTinh
+int SoLuong
+decimal DonGia
Ct tính toán SQL PERSISTED
+decimal ThanhTien «computed SQL PERSISTED»
ThanhToan
+int ThanhToanId [PK]
+string MaPhieuThu
+int KhachHangId [FK]
+DateTime NgayThanhToan
+decimal SoTien
+string PhuongThuc
+string? GhiChu
PhanBoThanhToan
+int PhanBoThanhToanId [PK]
+int ThanhToanId [FK]
+int HoaDonId [FK]
+string MaHoaDon
+decimal SoTienPhanBo
+DateTime NgayPhanBo
«DomainService»
ThuatToanFIFO
+PhanBo(IEnumerable<HoaDonNoModel> danhSach, decimal soTien, DateTime ngayThu) : KetQuaPhanBoFifo
«DomainService»
PhanTichTuoiNoService
+PhanBoVaoNhomTuoiNo(BaoCaoTuoiNoItem item, IEnumerable<HoaDonTuoiNoInput> hoaDons, DateTime ngayBaoCao) : void
+TinhSoNgayQuaHan(DateTime hanThanhToan, DateTime ngayBaoCao) : int
UseCases
(Ch tham chiu CoreBusiness)
Scoped Draft State Service
IHoaDonNhapStateService
+bool CoDuLieuNhap
+LayDuLieuNhap() : HoaDonNhapDto
+CapNhatNhap(int khachHangId, DateTime ngayLap, DateTime? hanThanhToan, string? ghiChu, IEnumerable<DongHangNhapDto> chiTiets) : void
+XoaNhap() : void
HoaDonNhapStateService
-HoaDonNhapDto _draft
+bool CoDuLieuNhap
+LayDuLieuNhap() : HoaDonNhapDto
+CapNhatNhap(int khachHangId, DateTime ngayLap, DateTime? hanThanhToan, string? ghiChu, IEnumerable<DongHangNhapDto> chiTiets) : void
+XoaNhap() : void
Use Cases iu Phi
ITaoHoaDonBanChiuUseCase
+ExecuteAsync(TaoHoaDonRequest request) : Task<TaoHoaDonResult>
+LayMaHoaDonTiepTheoAsync() : Task<string>
ITaoThanhToanUseCase
+ExecuteAsync(TaoThanhToanRequest request) : Task<TaoThanhToanResult>
+LayMaPhieuThuTiepTheoAsync() : Task<string>
IXuatBaoCaoTuoiNoExcelUseCase
+ExecuteAsync(XuatBaoCaoTuoiNoExcelRequest request) : Task<XuatBaoCaoTuoiNoExcelResult>
TaoHoaDonBanChiuUseCase
-readonly IHoaDonRepository _hoaDonRepository
-readonly ICurrentUserService _currentUserService
+ExecuteAsync(TaoHoaDonRequest request) : Task<TaoHoaDonResult>
+LayMaHoaDonTiepTheoAsync() : Task<string>
TaoThanhToanUseCase
-readonly IThanhToanRepository _thanhToanRepository
-readonly ICurrentUserService _currentUserService
+ExecuteAsync(TaoThanhToanRequest request) : Task<TaoThanhToanResult>
+LayMaPhieuThuTiepTheoAsync() : Task<string>
XuatBaoCaoTuoiNoExcelUseCase
-readonly IXemBaoCaoTuoiNoUseCase _xemBaoCaoTuoiNoUseCase
-readonly IBaoCaoTuoiNoExcelExporter _excelExporter
+ExecuteAsync(XuatBaoCaoTuoiNoExcelRequest request) : Task<XuatBaoCaoTuoiNoExcelResult>
gi interface
gi interface
gi interface
PluginInterfaces (nh ngha hp ng)
IHoaDonRepository
+LayDanhSachHoaDonAsync() : Task<IEnumerable<HoaDonListItemDto>>
+LayMaHoaDonTiepTheoAsync() : Task<string>
+TaoHoaDonBanChiuAsync(TaoHoaDonRequest request) : Task<TaoHoaDonResult>
+LayChiTietHoaDonAsync(int hoaDonId) : Task<ChiTietHoaDonDto?>
IBaoCaoTuoiNoRepository
+LayDuLieuBaoCaoAsync(DateTime ngayBaoCao) : Task<DuLieuBaoCaoTuoiNoDto>
+LayDuLieuTuoiNoRawAsync(DateTime ngayBaoCao) : Task<IEnumerable<KhachHangHoaDonTuoiNoRawDto>>
+LayTongTienThuChuaPhanBoAsync(DateTime ngayBaoCao) : Task<decimal>
IThanhToanRepository
+LayDanhSachThanhToanAsync() : Task<IEnumerable<ThanhToanListItemDto>>
+LayDanhSachHoaDonConNoTheoKhachHangAsync(int khachHangId) : Task<IEnumerable<HoaDonNoModel>>
+GhiNhanThanhToanAsync(TaoThanhToanRequest request) : Task<TaoThanhToanResult>
+LayMaPhieuThuTiepTheoAsync() : Task<string>
IBaoCaoTuoiNoExcelExporter
+ExportBaoCaoTuoiNo(BaoCaoTuoiNoResult baoCao) : byte[]
«implements»
«implements»
«implements»
«implements»
DataStore
SQL
(Tham chiu UseCases, CoreBusiness)
HoaDonRepository
-readonly ISqlConnectionFactory _connectionFactory
+LayDanhSachHoaDonAsync() : Task<IEnumerable<HoaDonListItemDto>>
+LayMaHoaDonTiepTheoAsync() : Task<string>
+TaoHoaDonBanChiuAsync(TaoHoaDonRequest request) : Task<TaoHoaDonResult>
+LayChiTietHoaDonAsync(int hoaDonId) : Task<ChiTietHoaDonDto?>
BaoCaoTuoiNoRepository
-readonly ISqlConnectionFactory _connectionFactory
+LayDuLieuBaoCaoAsync(DateTime ngayBaoCao) : Task<DuLieuBaoCaoTuoiNoDto>
+LayDuLieuTuoiNoRawAsync(DateTime ngayBaoCao) : Task<IEnumerable<KhachHangHoaDonTuoiNoRawDto>>
+LayTongTienThuChuaPhanBoAsync(DateTime ngayBaoCao) : Task<decimal>
ThanhToanRepository
-readonly ISqlConnectionFactory _connectionFactory
+LayDanhSachThanhToanAsync() : Task<IEnumerable<ThanhToanListItemDto>>
+LayDanhSachHoaDonConNoTheoKhachHangAsync(int khachHangId) : Task<IEnumerable<HoaDonNoModel>>
+GhiNhanThanhToanAsync(TaoThanhToanRequest request) : Task<TaoThanhToanResult>
+LayMaPhieuThuTiepTheoAsync() : Task<string>
Reports
Excel
(Tham chiu UseCases, CoreBusiness)
ClosedXmlBaoCaoTuoiNoExcelExporter
+ExportBaoCaoTuoiNo(BaoCaoTuoiNoResult baoCao) : byte[]
-SanitizeText(string? text) : string
Project Ph / Công c (Không nm trong lung chy chính ca web)
BanHang.InitTool
(Console App khi to CSDL)
BanHang.Tests
(xUnit Test Suite)

GHI CHÚ S RÚT GN (Architectural Summary):
S tp trung th hin các lp, interface tiêu biu ca 4 tng Clean Architecture
và gii thích Dependency Inversion Rule. Mt s DTO chi tit và phng thc ph tr
c lc bt m bo tính trc quan và kh nng c ca s .

PHÂN BIT 2 CHIU TRONG KIN TRÚC:
1. **Chiu ph thuc mã ngun (Compile-time Dependency Rule):**
   Tng Presentation & Plugins ..> Tng UseCases ..> Tng CoreBusiness (hng vào trong).
   CoreBusiness c lp 100%; UseCases không tham chiu các lp c th ca Plugins.
2. **Chiu gi thc thi lúc chy (Runtime Control Flow):**
   UI Page -> gi Interface UseCase -> UseCase gi Interface Repository
   -> Lp c th Repository (SQL Server / ClosedXML) thc thi truy vn.